

Michael Vinciguerra

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SUMMARY

Cybersecurity and Networking professional with hands-on experience in Ethernet protocol testing, network protocol analysis, traffic analysis, Python automation, cloud infrastructure, and threat detection systems. Experienced collaborating with engineers from leading technology companies including Cisco, Intel, AMD, and Apple. Strong background in networking, cybersecurity operations, and systems troubleshooting.

EXPERIENCE

University of New Hampshire Interoperability Labs (IOL)

February 2024 – June 2026

Network and Interoperability Engineer | Durham, NH

- Led IEEE 802.3 compliance and interoperability testing with global technology partners (AMD, Apple, Intel, Cisco, HP, Eero) with focused expertise in MAC layer testing (Clause 4) and Flow Control (Clause 31); configured and validated networking switches against standard requirements.
- Performed network protocol analysis using Wireshark, executed network troubleshooting and network validation across Layer 2 and Layer 3 environments, and conducted throughput and performance analysis with Spirent and Ixia.
- Served as primary point of contact during live vendor testing sessions with engineers from AMD, Apple, Intel, Cisco, HP, and Eero — diagnosed protocol failures in real time, communicated findings clearly across technical and non-technical stakeholders, and led post-session review meetings to present test results and remediation paths.
- Built a Python SNMP rack management system replacing a legacy C# application — multi-threaded Streamlit interface enabling real-time simultaneous monitoring and control of 12+ Triplite PDUs by all lab technicians.
- Developed a Python automation tool for device selection and formatted Word/Excel report generation, cutting lab setup time by 40%.
- Served as Reservation Manager for 1+ year, coordinating testing logistics for 20+ global vendor engagements — managed communications, device specification collection, and live on-call troubleshooting with vendor engineers, maintaining zero missed testing windows across all sessions.
- Mentored 4+ new hire technicians on IEEE standards, lab equipment, and debugging methodology — enabling all four to independently lead client testing sessions within their first month.

SKILLS

Machine Learning: scikit-learn, BERT, HuggingFace Transformers, RAG, Model Development

Security: SIEM (Splunk), Incident Response, Threat Hunting, Microsoft Defender, CrowdStrike, SOAR Workflows, Log Analysis, Security Monitoring, Incident Analysis

Cloud & Infrastructure: Microsoft Azure, Linux, Windows, macOS

Networking & Dev: Layer 2/3 Networking, Network Validation, Network Troubleshooting, TCP/IP, Ethernet, VLANs, Wireshark, IEEE 802.3, Spirent, Ixia, Git

Languages: Python, Java, C, SQL

PROJECTS

Urban Heat Island Predictive Modeling

August 2025 – December 2025

Machine Learning — University of New Hampshire

- Trained a Random Forest regression model ($R^2=0.747$, $RMSE=2.55^\circ C$) on a 19,800-point spatial grid using Landsat 8 thermal imagery, NYC building footprints, and street tree census data; managed the full ML pipeline from feature engineering through deployment.

- Simulated three climate warming scenarios and modeled green infrastructure interventions — achieving 1.28–1.90°C cooling reductions; identified impervious surfaces, NDVI, and vegetation score as top predictors.

Distributed Replicated Key-Value Store

August 2025 – December 2025

Cloud Computing — University of New Hampshire

- Designed and built a primary-backup replicated key-value store in Go with quorum-based write consistency, GOB-encoded RPC for inter-node replication, and an HTTP REST API; deployed across 4 Azure VMs on a custom VNet with private IP networking.
- Iterated through 4 optimization phases — diagnosing cloud RTT timeout misconfiguration, goroutine leaks, and unbounded log growth — achieving a 33x throughput improvement and 3.2x read gain via backup read offloading; benchmarked P50/P95/P99 latency across 1–3 concurrent clients.

Threat Detection & Analysis Learning Model

January 2025 – May 2026

Senior Capstone — UNH Cybersecurity Department

- Architected an AI-driven autonomous security analyst integrating Splunk SIEM for live event ingestion, a custom SOAR workflow for automated threat classification, and Ollama LLMs for structured incident response report generation — extending SOC coverage to 24/7.
- Built a retrieval-augmented generation (RAG) pipeline with vector database integration for context-aware threat hunting during active investigations; benchmarked multiple LLMs and SOAR platforms for model deployment in production; presented at UNH CS Poster Symposium, Spring 2026.

Steam Review Sentiment Analysis

January 2026 – May 2026

Machine Learning & NLP — University of New Hampshire

- Collected and processed 555K+ real-world Steam game reviews across 37 titles via the public API; built and compared multiple machine learning models for binary sentiment classification using Python and scikit-learn.
- Fine-tuned a BERT transformer model using HuggingFace Transformers, achieving over 99% classification accuracy and significantly outperforming all classical models, with the strongest gains on negative reviews.

EDUCATION

University of New Hampshire

May 2026

Bachelor of Science in Computer Science | Durham, NH

Relevant Coursework: Machine Learning, NLP, Cloud Computing, Cybersecurity Principles, Operating Systems, Data Structures & Algorithms, Database Systems & Design, Object-Oriented Design, Concurrency

LEADERSHIP

Pi Kappa Phi — Founding Father & Risk Manager

August 2022 – May 2026

University of New Hampshire

- Co-founded the chapter from the ground up, growing to 80+ active members; authored bylaws, organizational structure, and recruitment frameworks still in use today.
- Served as Risk Manager, developed and enforced chapter compliance policies and coordinated logistics for a formal chapter event.
- Mentored members on academic and professional development; contributed to organizing fundraising campaigns raising \$4,000+ for The Ability Experience.